KUMARAGURU COLLEGE OF TECHNOLOGY

DESIGN AND FABRICATION OF A 3D PRINTED WING FOR LOAD TEST

DEMONSTRATION
HACK A BIT WING DESIGN CHALLENGE REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

AERONAUTICAL ENGINEERING

KUMARAGURU COLLEGE OF TECHNOLOGY
COIMBATORE– 641049.

AERONAUTICAL ENGINEERING

BONAFIDE CERTIFICATE

Certified that this project report:
Design and Fabrication of a 3D Printed Wing for load Test Demonstration

Bonafide work of (Anuchand K N – 23BAE008, Jeevan Sk – 23BAE020, Manoj M – 23BAE024, Sharan Sundhar S – 23BAE046, Swathemica V – 23BAE054)
who carried out the project work under my supervision.

Internal Examiner

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The candidates with college roll/ register number 23BAE008, 23BAE020, 23BAE024, 23BAE046, 23BAE054 were examined in the Aircraft structures II Laboratory examination held

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Abstract

This project focuses on the design, fabrication, and testing of a wing prototype capable of withstanding bending loads while maintaining optimal structural efficiency. The study examines spar and rib placement to achieve balanced load distribution and strength-to-weight optimization. A NACA 2421 symmetric airfoil was modeled using SolidWorks, and the prototype was fabricated through FDM 3D printing using PLA material. The cantilever wing was tested under incremental loading, and deflection was measured using a dial gauge to assess stiffness and structural behavior. Results showed that the symmetric NACA 2421 airfoil provided excellent structural rigidity and stability, making it suitable for applications requiring balanced aerodynamic performance with minimal pitching moment. Although the symmetric profile produces lower lift compared to cambered airfoils, it offers superior structural strength and predictable behavior under varying load conditions. Future improvements include refining the internal spar geometry through computational optimization and integrating composite materials to enhance overall performance and durability.

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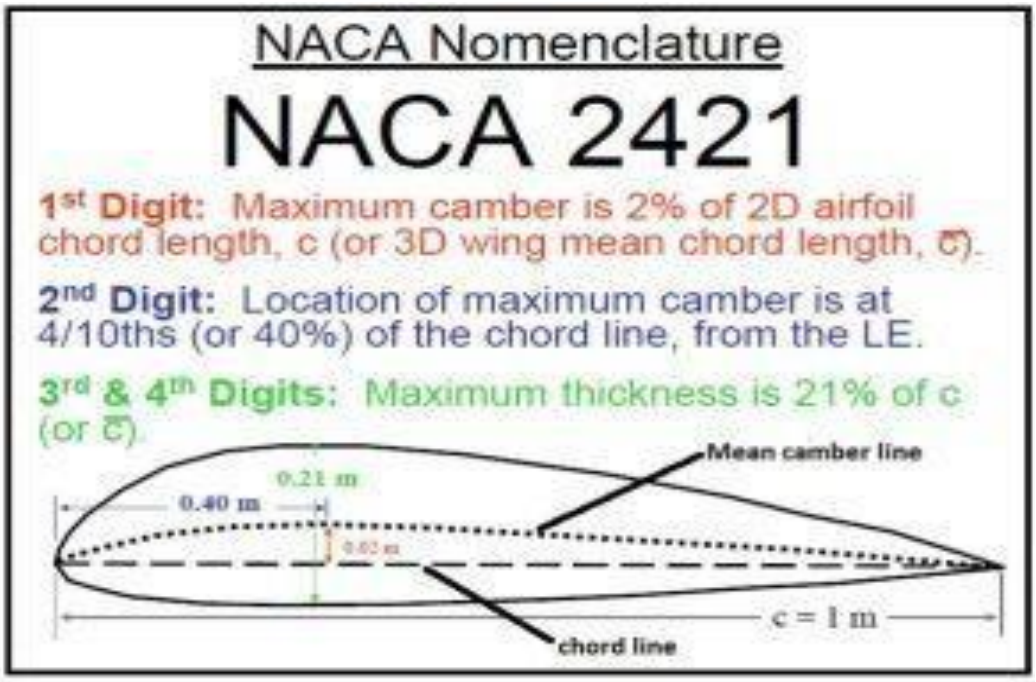
1.INTODUCTION:

1. The development of lightweight and high-strength structures has become a major focus in modern aerospace engineering. Aircraft wings are designed to provide lift while maintaining structural integrity under aerodynamic and load conditions. Additive manufacturing (3D printing) offers new possibilities for rapid prototyping and experimental testing of airfoil models.

2. This project, titled "**Design and Fabrication of a 3D Printed Wing for load Test Demonstration**. The study aims to evaluate the performance of the airfoil under different load conditions to analyze its stiffness and deflection behavior.

3. A NACA 2421 airfoil profile was selected due to its symmetric geometry, high stiffness, and strength characteristics. Using SolidWorks, the 3D model of the wing including ribs and spars was created, and ANSYS software was used to analyze the structural performance under various load conditions. The physical prototype was fabricated using PLA filament through 3D printing for experimental validation.



1. **Design Thinking Process:**
2. **Empathize: Researched various aircraft wing configurations to gain insight into their strength characteristics, weight distribution, and load response.**
3. **Define: Selected a cantilever wing design featuring a rectangular spar to enhance load-carrying ability while keeping the overall structure lightweight.**
4. **Ideate: Evaluated different spar shapes and materials, emphasizing aerodynamic efficiency, structural strength, and manufacturability.**
5. **Prototype: Created a CAD model of the spar and initiated 3D printing using lightweight materials to enable precise physical testing.**
6. **Test: Plan to subject the printed spar to gradually increasing loads to study its deformation behaviour, detect failure zones, and assess its structural stability under stress**
7. **Airfoil Design & CAD Modeling:**

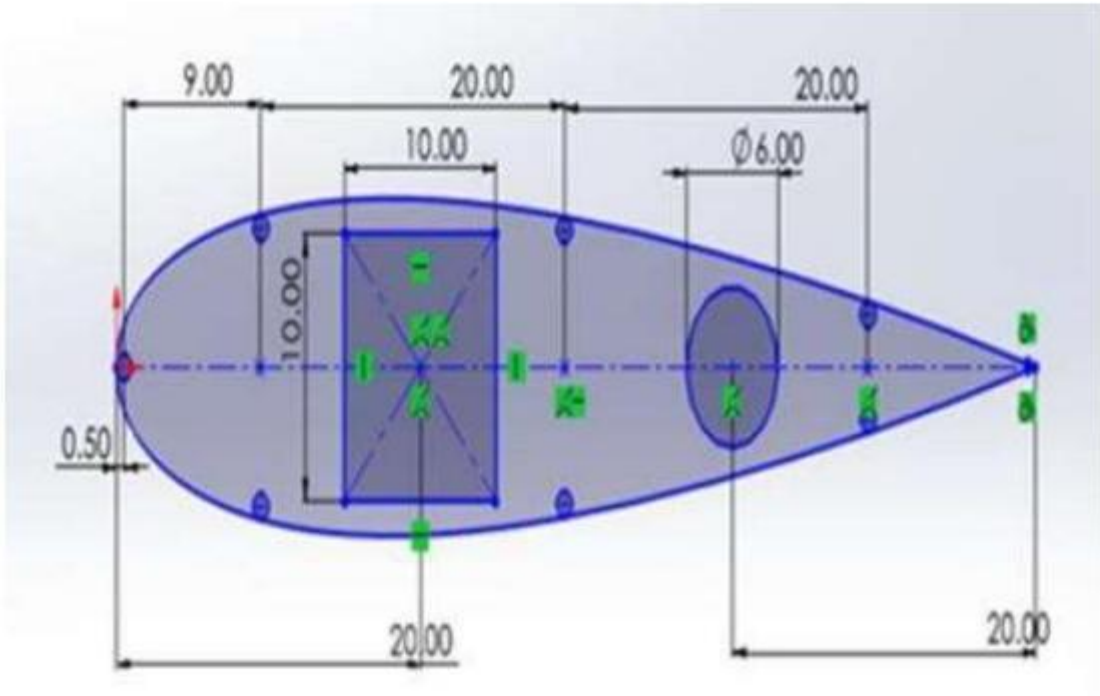
The NACA 2421 airfoil was selected for this project due to its symmetric shape, which provides balanced aerodynamic characteristics and high structural strength. The model was created using SolidWorks software for precise 3D design and analysis.

Steps Involved:

Select Airfoil: Selected the NACA 2421 airfoil as the base profile for the wing design.
Sketch Profile: Created the 2D outline of the airfoil in SolidWorks using the standard NACA 2421 coordinate data.
Extrude Model: Extended the 2D airfoil sketch along the required span to generate the 3D wing model.

Add Structural Elements:

Spar: 60.54 mm length, 8.50 mm breadth, and 5.00 mm depth
Rib: Chord length 60.54 mm, maximum height 12.94 mm
I-Section: 10 mm × 10 mm overall, with 3 mm flange thickness and 4 mm web thickness



5. **Assemble Parts**: Ribs and spar were positioned correctly to form the complete wing structure.

6. **Export File**: The final model was exported as an STL file for 3D printing and further analysis.

The CAD model ensured proper alignment, strength, and lightweight design, making it suitable for fabrication and load testing.

2.2Fabrication process :

1. Printing Parameters:

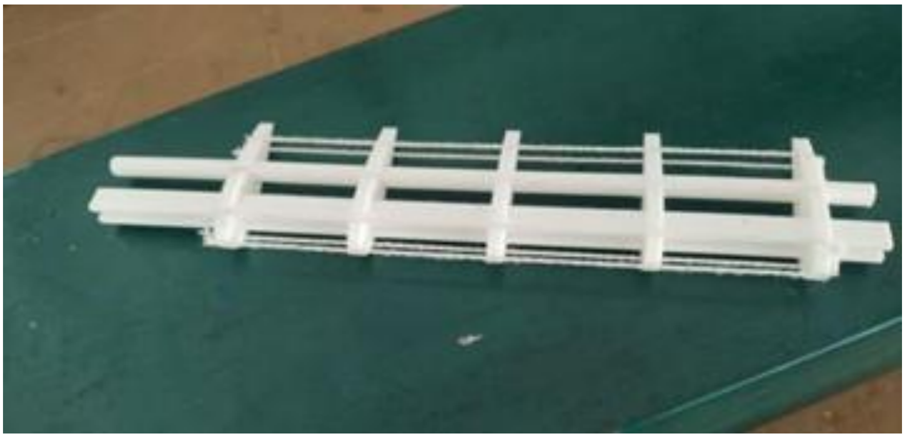
2. Nozzle: 0.4 mm

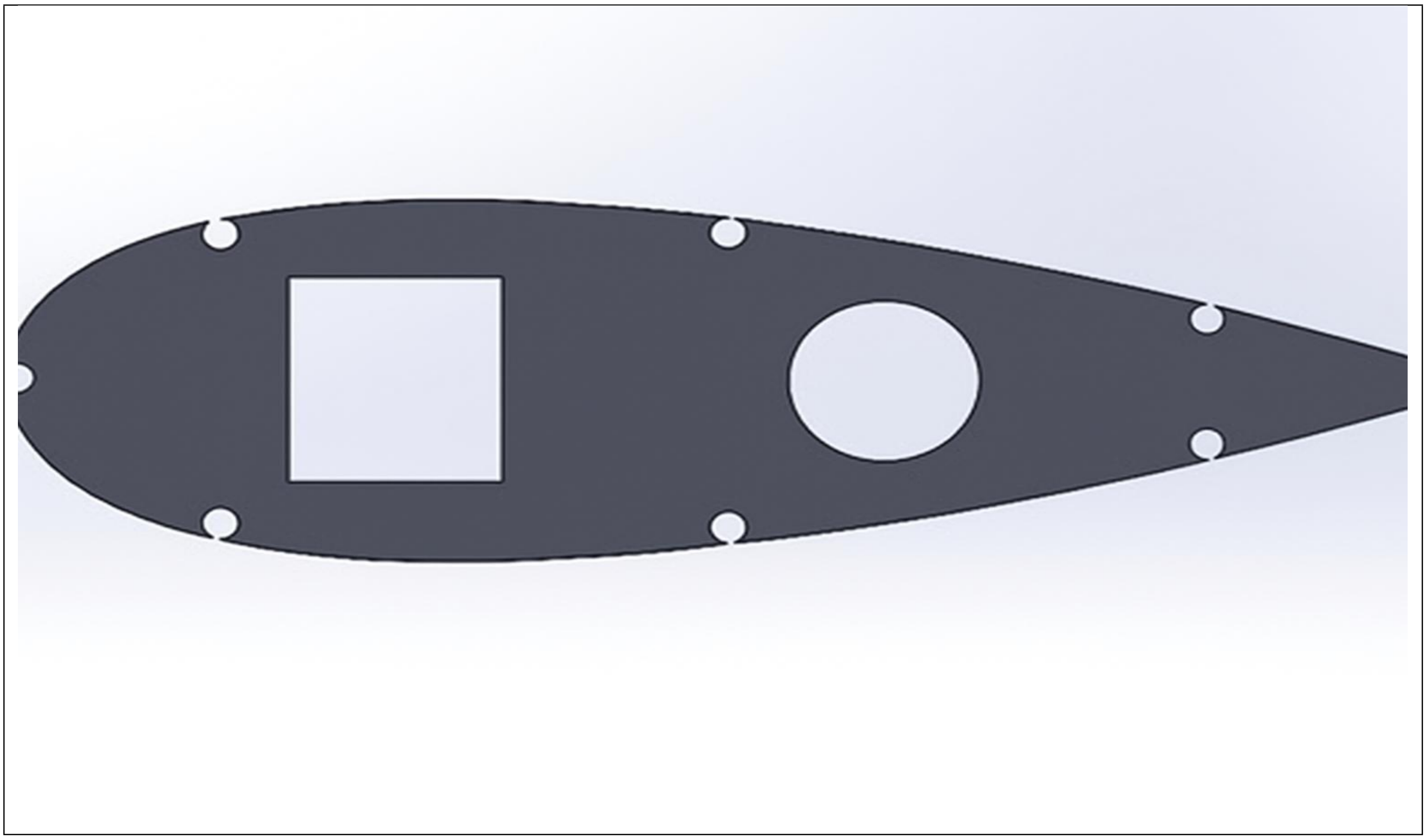
3. Layer height: 0.2 mm

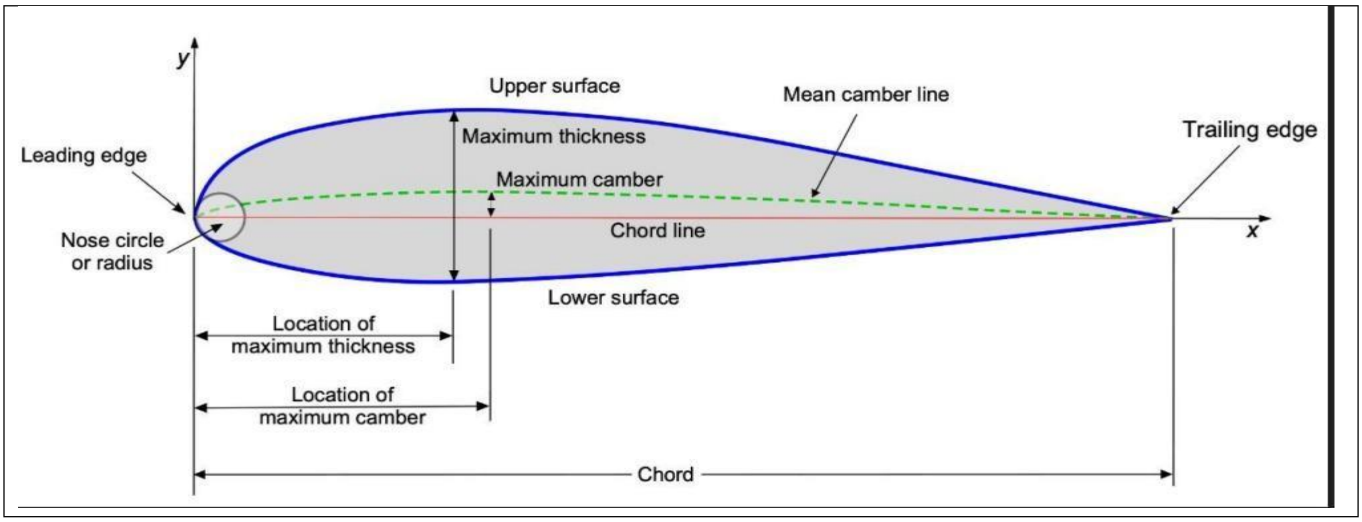
4. Infill: 20%

5. Material used: 80 g PLA

6. Print time: 98 minutes






y
Upper surface
Mean camber line
Maximum thickness
Trailing edge
Leading edge
Maximum camber
Chord line
x
Nose circle
or radius
Lower surface
Location of
maximum thickness
Location of
maximum camber
Chord

TEST SETUP :

- The wing was clamped securely at one end to act as a fixed support.
- Incremental loads of 5 N, 10 N, and 15 N were applied at the free end.
- The tip deflection was measured for each load using a scale or dial gauge.
- Readings were noted carefully to maintain accuracy.
- The

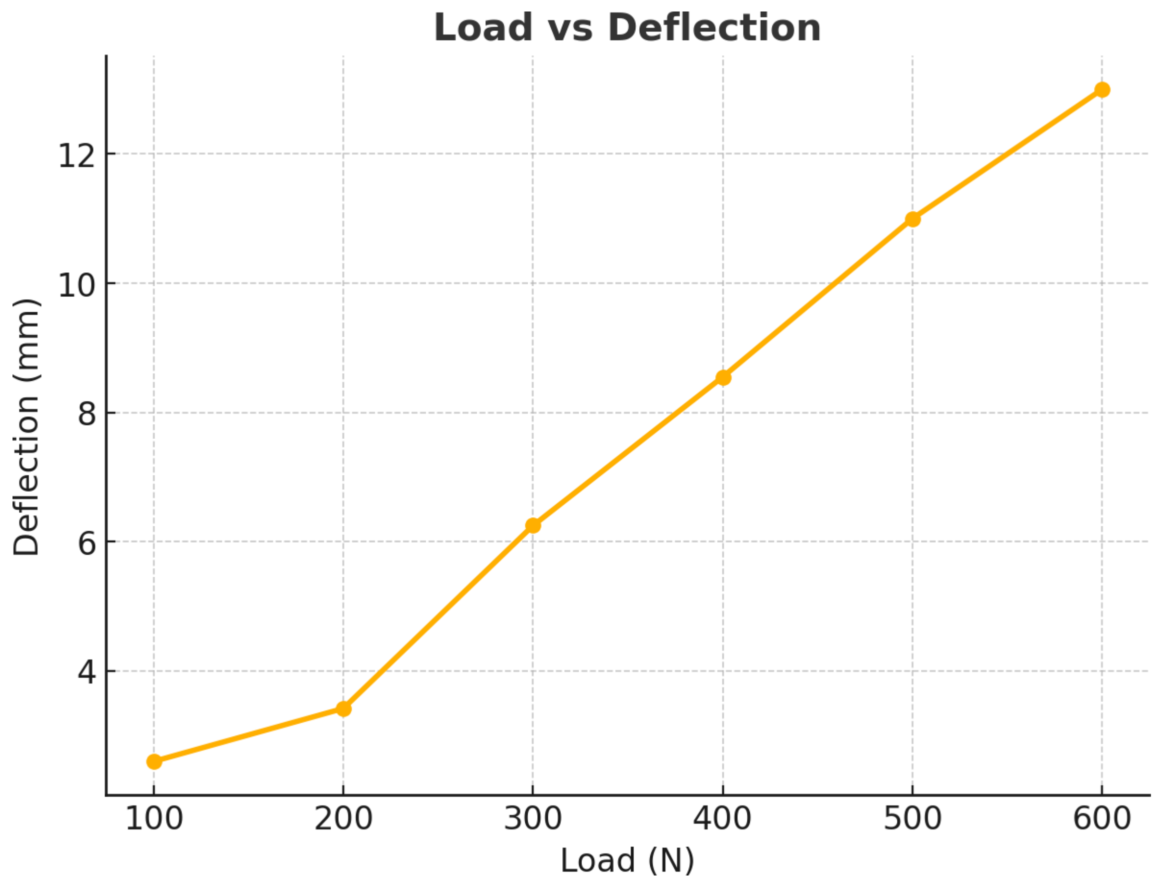
TABULATION :

Load	Deflection (mm)	Moment of Inertia
100	2.6	7.20x10^4
200	3.42	7.48x10^4
300	6.25	7.72x10^4
400	8.55	8.77x10^4
500	11	10.28x10^4
600	13	8.20x10^4 (weak)
700		Breaked

RESULTS :

Parameter	Value/obser vation
Load	300 g
Deflection	6,25

Graph :



Load

3- <u>Conclusion:</u>

The NACA 4415 airfoil was designed in CATIA and 3D printed successfully. Test results showed a linear load–deflection relationship, confirming elastic behavior and good structural strength.

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